

DIGITAL SAT — MATH

Theory Companion — Part 2

Advanced Math & Geometry / Trigonometry — Theory & Worked Problems

- 16 skills, one representative Hard/Very Hard worked problem per skill
- Advanced Math: Functions · Composite & Inverse · Transformations · Polynomials & Factoring · Poly Division & Rational Expressions · Quadratics · Radicals & Exponents · Exponential Functions · Radical & Rational Equations
- Geometry and Trigonometry: Angles & Triangles · Right Triangles & Trig · Radians & the Unit Circle · Circles · Volume & Surface Area
- Companion to the full 16-skill, 128-question Math Theory Study Companion — Greek terminology included for every skill

The 16 Skills

1. Functions — Notation, Domain, Range & Graphs
2. Composite & Inverse Functions
3. Function Transformations
4. Polynomial Operations & Special Products
5. Factoring Polynomials
6. Dividing Polynomials & Rational Expressions
7. Solving Quadratic Equations & the Discriminant
8. Quadratic Graphs, Roots & Word Problems
9. Radicals & Rational Exponents
10. Exponential Functions — Growth & Decay
11. Radical & Rational Equations
12. Angles, Parallel Lines & Triangle Fundamentals
13. Right Triangles & Trigonometry
14. Radians & the Unit Circle
15. Circles — Circumference, Arcs, Sectors, Equations & Theorems
16. Volume, Surface Area & Similarity of Solids



PART 2

Geometry and Trigonometry

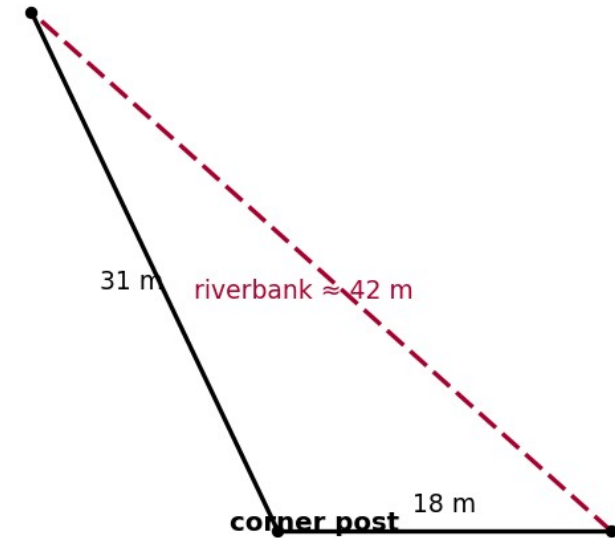
Angles, Parallel Lines & Triangle Fundamentals • Right Triangles & Trigonometry • Radians & the Unit Circle • Circles — Circumference, Arcs, Sectors, Equations & Theorems • Volume, Surface Area & Similarity of Solids

Core Rules — In Detail

- Complementary & Supplementary Angles — Complementary: sum = 90° . Supplementary: sum = 180° . Vertical angles (formed by two intersecting lines) are always equal; adjacent angles share a vertex and a side.
- Parallel Lines Cut by a Transversal — Corresponding, alternate interior, and alternate exterior angle pairs are EQUAL. Same-side interior angle pairs are SUPPLEMENTARY (sum to 180°).
- Triangle Angle Sum — $A + B + C = 180^\circ$ True for every triangle, regardless of shape — knowing any two angles gives you the third instantly.
- Classifying Triangles — By angle: acute (all $< 90^\circ$), right (one $= 90^\circ$), obtuse (one $> 90^\circ$). By side: equilateral (all sides equal), isosceles (two sides equal, base angles equal), scalene (no sides equal).
- Triangle Inequality Theorem — $a + b > c$ (for all three pairs of sides) The sum of any two sides must exceed the third side, or no triangle can be formed — use this to bound a missing side's possible range.
- Side-Angle Relationship — The largest side is always opposite the largest angle, and the smallest side is opposite the smallest angle.
- Exterior Angle Theorem — Exterior angle = sum of the two remote interior angles Formed by extending one side of a triangle; equals the sum of the two angles that are NOT adjacent to it.

Watch Out: Same-side interior angles are the one pair in the parallel-lines diagram that are SUPPLEMENTARY, not equal — mixing this up with corresponding/alternate angles is the most common error. ⚠ The triangle inequality gives you a RANGE ($5 < x < 19$, for example), not a single boundary — check that a candidate side length satisfies BOTH the upper and lower bound.

A landscaping company is fencing a triangular plot of land next to a river in Utrecht. Two sides of the triangle are already fixed: one fence runs 18 meters and the other runs 31 meters, meeting at a shared corner post. The third side runs along the riverbank (unfenced, since the river itself forms the boundary), but for cost-estimation purposes the company treats it as if it were fenced too. The riverside length must be a whole number of meters, and the total perimeter of the plot (all three sides added together) must be a multiple of 7 meters. What is the greatest possible perimeter satisfying all of these conditions?



Formula / Rule Used: Triangle Inequality Theorem: the third side must be strictly between the difference and the sum of the other two sides.

Work through it, then advance for the full step-by-step solution.



Step 1: Let x = the riverside length. By the Triangle Inequality Theorem, x must satisfy $31 - 18 < x < 31 + 18$.

Step 2: Simplify: $13 < x < 49$.

Step 3: The perimeter is $P = 18 + 31 + x = 49 + x$, so shift the inequality by adding 49 to all three parts: $13 + 49 < P < 49 + 49$.

Step 4: This gives $62 < P < 98$.

Step 5: List the multiples of 7 strictly between 62 and 98: 63, 70, 77, 84, 91.

Step 6: The greatest of these is 91.

Step 7: Check: this corresponds to $x = 91 - 49 = 42$, and $13 < 42 < 49$ is true, so a valid triangle exists.

Answer: 91 meters (with riverside length 42 meters)

Tip: For a triangle-inequality range problem, find the valid range for the unknown side first, then shift that same range by the known sides' sum to get the perimeter's range directly — don't recompute the perimeter range from scratch.

Trap: The inequality $13 < x < 49$ is strict (not 'less than or equal to'), so a value at either boundary produces a degenerate triangle (a straight line, not a real triangle) — using $x = 49$ or $x = 13$ anyway is the specific error this problem punishes.

Tempting Wrong Answers — and Exactly What Went Wrong

98 meters

Used $x = 49$, the excluded upper boundary of the range, forgetting the inequality is strict, which actually produces a degenerate (flat) triangle, not a valid one.

63 meters

Found the correct range $62 < P < 98$ but picked the SMALLEST valid multiple of 7 in that range instead of the greatest, misreading 'greatest possible perimeter' as 'least'.

90 meters

Correctly narrowed down to multiples of 7 below 98 but mistakenly selected 90 as the answer, which isn't actually a multiple of 7 ($90/7$ is not a whole number).

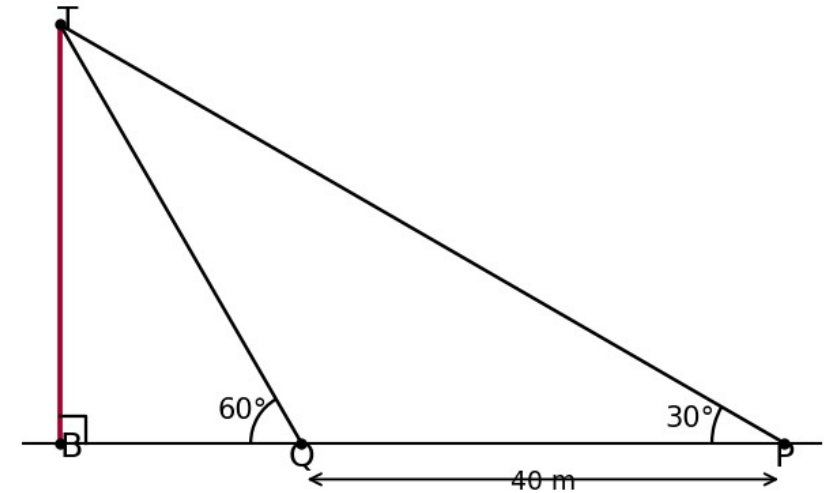
Shortcut: Shift the side-length inequality ($13 < x < 49$) by adding 49 to every part to get the perimeter's range directly ($62 < P < 98$), skipping a separate perimeter recalculation.

Core Rules — In Detail

- The Pythagorean Theorem — $a^2 + b^2 = c^2$ c is always the hypotenuse — the side opposite the right angle, and the longest side of the triangle.
- Pythagorean Triples — 3-4-5, 5-12-13, 8-15-17, 7-24-25 (and any whole-number multiple) Recognizing a scaled triple (e.g. 15-36-39 = $3 \times (5-12-13)$) instantly gives you the third side without computing any squares.
- 45-45-90 Triangle — Legs: x, x . Hypotenuse: $x\sqrt{2}$ An isosceles right triangle — both legs equal, and the hypotenuse is always $\sqrt{2}$ times a leg.
- 30-60-90 Triangle — Sides: $x, x\sqrt{3}, 2x$ (opposite $30^\circ, 60^\circ, 90^\circ$ respectively) The side opposite 30° is the shortest; the hypotenuse (opposite 90°) is always twice that shortest side.
- Similar Triangles — AA, SSS, or SAS similarity Similar triangles have equal corresponding angles and proportional corresponding sides — set up a proportion between matching sides to solve for an unknown.
- Congruent Triangles — SSS, SAS, ASA, AAS, or HL (right triangles only) Congruent triangles are identical in every corresponding side and angle.
- SOH-CAH-TOA — $\sin\theta = \text{Opp}/\text{Hyp}$, $\cos\theta = \text{Adj}/\text{Hyp}$, $\tan\theta = \text{Opp}/\text{Adj}$ Identify which given/unknown side is opposite vs. adjacent to the angle in question before choosing a ratio.
- Complementary Angle & Pythagorean Identities — $\sin(90^\circ - \theta) = \cos\theta$. $\sin^2\theta + \cos^2\theta = 1$. The two acute angles of any right triangle are complementary, which makes sine and cosine of complementary angles equal.

Watch Out: Trigonometric ratios depend ONLY on the angle, never on the triangle's size — this is exactly why they're preserved across similar triangles. ⚠ In a 30-60-90 triangle, the side opposite 60° is $x\sqrt{3}$, NOT $2x$ — confusing which side pairs with which angle is the most common special-triangle error.

A surveyor in the town of Meteora stands at point P on flat ground and measures the angle of elevation to the top of a rock pillar, point T, as 30° . She then walks 40 meters directly toward the base of the pillar, point B (with P, her new position Q, and B all on the same straight line), and measures the new angle of elevation to T from Q as 60° . Points P, Q, and B are all at the same ground elevation, and angle TBP is a right angle. What is the height of the pillar, BT, in meters?



Formula / Rule Used: 30-60-90 triangle side ratios combined with setting up two right triangles that share the same height.

Work through it, then advance for the full step-by-step solution.

Step 1: Let $h = BT$ (the height of the pillar) and let $x = QB$ (the horizontal distance from the second position to the base).

Step 2: In right triangle TBQ , angle $TQB = 60^\circ$, so this is a 30-60-90 triangle with the side opposite 60° (which is h) related to the side opposite 30° ... instead use tangent directly: $\tan(60^\circ) = h/x$, so $h = x \cdot \sqrt{3}$.

Step 3: In right triangle TBP , angle $TPB = 30^\circ$, and $PB = PQ + QB = 40 + x$.

Step 4: Apply tangent again: $\tan(30^\circ) = h/(40 + x)$, so $h = (40 + x) \cdot (1/\sqrt{3})$.

Step 5: Set the two expressions for h equal: $x \cdot \sqrt{3} = (40 + x)/\sqrt{3}$.

Step 6: Multiply both sides by $\sqrt{3}$: $3x = 40 + x$.

Step 7: Solve for x : $2x = 40$, so $x = 20$.

Step 8: Substitute back into $h = x \cdot \sqrt{3}$: $h = 20 \cdot \sqrt{3}$ meters (approximately 34.6 meters).

Answer: $BT = 20 \cdot \sqrt{3}$ meters (approximately 34.6 m)

Tip: For two-station angle-of-elevation problems, define the unknown height and the unknown near-distance as two variables, write tangent equations for both triangles, and solve the resulting system — don't try to jump straight to the answer with one triangle.

Trap: A student often tries to use only the far triangle (angle 30° , distance 40) and solves $h = 40 \cdot \tan(30^\circ) = 40/\sqrt{3}$, ignoring that 40 is NOT the full distance from P to B (it's only the distance she walked, not PB), producing a wrong height.

Tempting Wrong Answers — and Exactly What Went Wrong

BT = $40/\sqrt{3}$ meters (approx 23.1 m)

Incorrectly treated the 40-meter walk as the full distance PB from the original point to the base, using $\tan(30^\circ) = h/40$ instead of setting up the two-triangle system.

BT = $40\sqrt{3}$ meters (approx 69.3 m)

Incorrectly treated the 40-meter walk as QB (the distance from the second position to the base) and used $\tan(60^\circ) = h/40$ directly, skipping the system entirely.

BT = 20 meters

Solved the system correctly to find $x = 20$ but then reported x itself as the final answer instead of substituting back to find $h = x\sqrt{3}$.

Shortcut: Once you notice the two angles are 30° and 60° (complementary-looking but really just both present), the ratio $\tan(60^\circ)/\tan(30^\circ) = 3$ tells you PB is exactly 3 times QB, so $x = 40/(3-1) = 20$ in one line without full algebra.

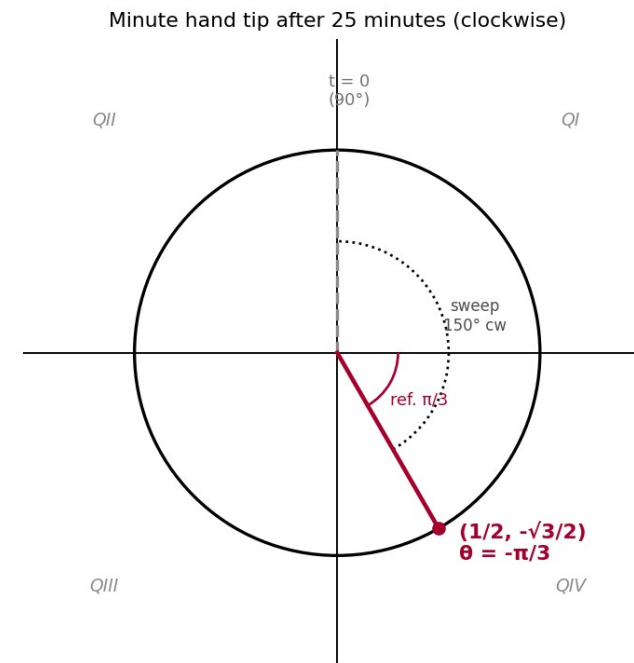
Desmos Tip: Graph the two lines $y = x \cdot \tan(60^\circ)$ and $y = (x+40) \cdot \tan(30^\circ)$ using x as the horizontal distance from Q, and find their intersection point's y -value to confirm $h = 20\sqrt{3}$.

Core Rules — In Detail

- Degree-Radian Conversion — π radians = 180° Multiply degrees by $\pi/180$ to get radians; multiply radians by $180/\pi$ to get degrees.
- Common Angles in Radians — $30^\circ=\pi/6$, $45^\circ=\pi/4$, $60^\circ=\pi/3$, $90^\circ=\pi/2$, $180^\circ=\pi$, $360^\circ=2\pi$ Memorizing this table lets you convert instantly without setting up a proportion each time.
- The Unit Circle — Point at angle $\theta = (\cos\theta, \sin\theta)$ A circle of radius 1 centered at the origin — the x-coordinate of the point at angle θ is always $\cos\theta$, and the y-coordinate is always $\sin\theta$.
- Quadrant Signs — QI: (+,+) QII: (-,+) QIII: (-,-) QIV: (+,-) Use the reference angle (the acute angle to the x-axis) to find the magnitude, then apply the correct sign for the quadrant.
- Coterminal Angles — Add or subtract 360° (or 2π) repeatedly Coterminal angles point in the same direction and share the same trig values; use this to bring any angle into the standard $[0^\circ, 360^\circ)$ or $[0, 2\pi)$ range.

Watch Out: A negative angle or an angle over 360° is not 'wrong' — just add or subtract full rotations (360° or 2π) until it lands in the standard range, then read off the trig values normally. ⚠ The sign of cosine and sine depends entirely on the quadrant — always determine the quadrant before assigning + or - to the reference-angle value.

A large decorative clock in a plaza in Vienna has a minute hand modeled as a point moving on a unit circle (radius 1) centered at the clock's center. At time $t = 0$ minutes, the tip of the minute hand is at the top of the clock, corresponding to standard position angle $\pi/2$. The minute hand moves clockwise at a constant rate, completing one full revolution (2π radians) every 60 minutes. What are the exact coordinates of the tip of the minute hand at $t = 25$ minutes?



Formula / Rule Used: Angular rate = 2π / period; clockwise rotation subtracts angle from the starting position.

Work through it, then advance for the full step-by-step solution.



Step 1: Find the rotation rate: one full revolution (2π radians) every 60 minutes gives $2\pi/60 = \pi/30$ radians per minute.

Step 2: Find the angle swept in 25 minutes: $25 \times \pi/30 = 25\pi/30 = 5\pi/6$ radians.

Step 3: Since the hand moves clockwise, subtract this swept angle from the starting angle instead of adding it: new angle = $\pi/2 - 5\pi/6$.

Step 4: Use a common denominator of 6: $\pi/2 = 3\pi/6$, so new angle = $3\pi/6 - 5\pi/6 = -2\pi/6 = -\pi/3$.

Step 5: The angle $-\pi/3$ has reference angle $\pi/3$ and lies in Quadrant IV (right of the y-axis, below the x-axis).

Step 6: In Quadrant IV, cosine is positive and sine is negative: $\cos(-\pi/3) = \cos(\pi/3) = \frac{1}{2}$, and $\sin(-\pi/3) = -\sin(\pi/3) = -\sqrt{3}/2$.

Step 7: So the tip of the minute hand is at $(\frac{1}{2}, -\sqrt{3}/2)$.

Answer: $(\frac{1}{2}, -\sqrt{3}/2)$

Tip: When a rotation is described as clockwise, subtract the swept angle from the starting angle instead of adding — direction sets the sign before you touch magnitude.

Trap: A rushed student adds $(5\pi)/6$ to $\pi/2$ instead of subtracting (forgetting that 'clockwise' reverses the usual counterclockwise-positive convention), landing at $(4\pi)/3$ and getting $(-\frac{1}{2}, -\sqrt{3}/2)$ instead of the correct $(\frac{1}{2}, -\sqrt{3}/2)$.

Tempting Wrong Answers — and Exactly What Went Wrong

$(-\frac{1}{2}), -\sqrt{\frac{3}{2}}$

Added the swept angle $5\pi/6$ to the start $\pi/2$ instead of subtracting for clockwise motion, landing at $4\pi/3$ instead of $-\pi/3$.

$(-\frac{1}{2}), \sqrt{\frac{3}{2}}$

Correctly subtracted to get $-\pi/3$ but then applied Quadrant II sign rules (cos negative, sin positive) instead of Quadrant IV, flipping both signs.

$(\frac{1}{2}), \sqrt{\frac{3}{2}}$

Found the correct reference angle $\pi/3$ but forgot the hand is below the x-axis at $t = 25$ min, so left sine positive instead of negative.

Shortcut: Since $25/60 = 5/12$ of a full revolution, jump straight to $5/12 \times 360^\circ = 150^\circ$ clockwise from 90° , i.e. an angle of -60° , without ever computing the $\pi/30$ -per-minute rate explicitly.

Desmos Tip: Type the point $(\cos(-\pi/3), \sin(-\pi/3))$ into the calculator in radian mode to confirm it evaluates to $(0.5, -0.866\dots)$.

Core Rules — In Detail

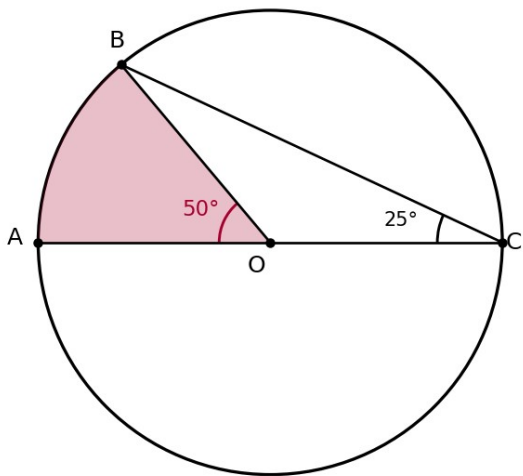
- Circumference and Area — $C = 2\pi r = \pi d$. $A = \pi r^2$ Diameter is twice the radius; both formulas depend only on r (or d).
- Arc Length — Arc length = $(\theta/360^\circ) \times 2\pi r$ An arc is exactly that same fraction of the full circumference that its central angle θ is of 360° .
- Sector Area — Sector area = $(\theta/360^\circ) \times \pi r^2$ Same proportional logic as arc length, applied to the full circle's area instead of its circumference.
- Equation of a Circle — $(x - h)^2 + (y - k)^2 = r^2$ Center is at (h, k) — note both signs are subtracted even though the center coordinates might be negative (e.g. center $(-3, 2)$ still uses $(x - (-3))^2 = (x + 3)^2$).
- Completing the Square for a Circle — Group x -terms and y -terms separately, complete the square for each (adding the same amount to both sides), then read off the center and radius from the resulting standard form.
- Tangent Lines — Radius $\perp$ tangent line A tangent touches the circle at exactly one point, and the radius drawn to that point is always perpendicular to the tangent — this creates a right triangle you can solve with the Pythagorean theorem.
- Inscribed Angle Theorem — Inscribed angle = $(1/2) \times$ intercepted arc An angle inscribed in a semicircle (intercepting a diameter) is always exactly 90° ; two inscribed angles that intercept the same arc are equal to each other.

Watch Out: In the circle equation, the center's y -coordinate sign gets flipped inside the formula: a center at (h, k) still appears as $(x - h)^2 + (y - k)^2$ even when h or k is negative. ⚠ Arc length and sector area both use $\theta/360^\circ$ as the fraction — using the arc's chord length or any other measurement instead of the central angle is a common setup error.

Worked Problem · Very Hard

Circle O has a radius of 18 meters, and segment AC is a diameter of circle O. Points A, B, and C all lie on circle O, and the inscribed angle ACB (with vertex C, intercepting the minor arc AB that does not contain C) measures 25° . What is the length of minor arc AB, in meters, in terms of π ?

Circle O, $r = 18$ m — inscribed $\angle ACB = 25^\circ$



arc AB = 5π meters

Formula / Rule Used: Inscribed Angle Theorem combined with the Arc Length formula.

Work through it, then advance for the full step-by-step solution.

Step 1: By the Inscribed Angle Theorem, an inscribed angle is half the measure of the arc it intercepts, so the central angle AOB is double the inscribed angle ACB.

Step 2: Central angle AOB = $2(25^\circ) = 50^\circ$.

Step 3: (Note: AC being a diameter is extra information here — it is not needed to find arc AB, since angle ACB already gives the intercepted arc directly.)

Step 4: Apply the arc length formula: arc length = $(\theta/360^\circ) \times 2\pi r$.

Step 5: Substitute $\theta = 50^\circ$ and $r = 18$: arc length = $(50/360) \times 2\pi(18) = (50/360) \times 36\pi$.

Step 6: Simplify: $(50 \times 36)/360 = 1800/360 = 5$.

Step 7: Arc length AB = 5π meters.

Answer: Arc AB = 5π meters

Tip: Whenever an inscribed angle is given, always double it first to get the central angle before touching the arc length formula — the inscribed angle itself is never the θ you plug in.

Trap: A rushed student plugs the inscribed angle (25°) directly into the arc length formula as if it were the central angle, skipping the required doubling step from the Inscribed Angle Theorem.



Tempting Wrong Answers — and Exactly What Went Wrong

2.5π meters

Used the inscribed angle (25°) directly as the central angle instead of doubling it per the Inscribed Angle Theorem, computing $(25/360) \times 2\pi(18) = 2.5\pi$ instead of 5π .

10π meters

Mistook the given radius (18) for a diameter and doubled it to 36 before applying the arc length formula, instead of using 18 as the radius directly.

50

Correctly doubled the inscribed angle to find the central angle (50°) but stopped there, reporting the angle measure itself instead of converting it into an arc length.

Shortcut: Combine the doubling and the arc length fraction into one line — $\text{arc} = (2 \times 25 / 360) \times 2\pi(18)$ — instead of writing the central angle as a separate step.

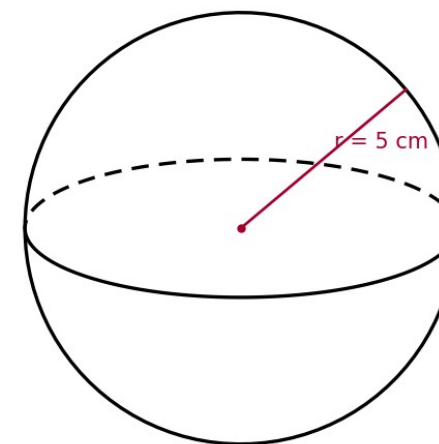
Core Rules — In Detail

- Prisms and Cylinders — $V = (\text{area of base}) \times \text{height}$. Rectangular prism: $V=lwh$. Cylinder: $V=\pi r^2 h$ Any solid with two parallel, congruent bases and straight sides uses this same 'base times height' logic.
- Pyramids and Cones — $V = (1/3) \times (\text{area of base}) \times \text{height}$. Cone: $V=(1/3)\pi r^2 h$. Square pyramid: $V=(1/3)s^2 h$ A pyramid or cone always holds exactly 1/3 the volume of a prism/cylinder with the same base and height.
- Spheres — $V = (4/3)\pi r^3$. Surface area = $4\pi r^2$ Both formulas depend only on the radius.
- Surface Area of Common Solids — Cube: $S=6s^2$. Rect. prism: $S=2(ab+bc+ac)$. Cylinder: $S=2\pi rh+2\pi r^2$ Surface area adds up the area of every face (or the lateral surface plus the two circular bases, for a cylinder).
- Composite Solids — Break the solid into recognizable pieces (e.g. a cylinder plus a hemisphere), find each volume separately, and add (or subtract, for a hollowed-out shape).
- Polygon Interior Angle Sum — $(n - 2) \times 180^\circ$ n is the number of sides; a triangle ($n=3$) gives 180° , a quadrilateral ($n=4$) gives 360° , and so on.
- Similarity Ratios — Perimeter ratio = λ . Area ratio = λ^2 . Volume ratio = λ^3 If two shapes are similar with scale factor λ , every LENGTH scales by λ , every AREA scales by λ^2 , and every VOLUME scales by λ^3 — this lets you jump between volume, surface area, and length ratios without finding actual dimensions.

Watch Out: Squaring or cubing the scale factor (not the raw dimension) is the key move for similarity problems — a volume ratio of 8 means a LENGTH ratio of 2 (since $2^3=8$), not a length ratio of 8. ⚠ A hemisphere's volume is exactly HALF a full sphere's — remember to divide $(4/3)\pi r^3$ by 2, not just reuse the sphere formula directly, in a composite-solid problem.

A spherical marker buoy manufactured by a shipping company in Bergen has a total surface area of 314 square centimeters. Using $\pi \approx 3.14$, what is the volume of the buoy, in cubic centimeters, rounded to the nearest whole number?

Spherical Buoy



$$\text{Surface Area} = 4\pi r^2 = 314 \text{ cm}^2$$

Formula / Rule Used: Sphere surface area $SA = 4\pi r^2$ solved for r , then substituted into sphere volume $V = (\frac{4}{3})\pi r^3$.

Work through it, then advance for the full step-by-step solution.

Step 1: Set up the surface area equation: $4\pi r^2 = 314$.

Step 2: Substitute $\pi \approx 3.14$: $4(3.14)r^2 = 314$, so $12.56r^2 = 314$.

Step 3: Solve for r^2 : $r^2 = 314/12.56 = 25$.

Step 4: Take the square root to find r : $r = \sqrt{25} = 5$ centimeters.

Step 5: Substitute $r = 5$ into the volume formula: $V = (\frac{4}{3})\pi r^3 = (\frac{4}{3})(3.14)(5^3) = (\frac{4}{3})(3.14)(125)$.

Step 6: Compute: $(\frac{4}{3})(3.14)(125) = (\frac{4}{3})(392.5) = 523.33$, which rounds to 523 cubic centimeters.

Answer: 523 cubic centimeters

Tip: When surface area is given but volume is asked for, always solve for the radius first (undoing the square), then plug that radius into the volume formula — never skip straight from the SA equation to a volume calculation.

Trap: A student who solves $r^2 = 25$ but forgets to take the square root often plugs $r^2 = 25$ directly into the volume formula as if it were r itself, producing a wildly inflated answer.

Tempting Wrong Answers — and Exactly What Went Wrong

65,417 cubic centimeters

Solved $r^2 = 25$ correctly but skipped the square root step, substituting 25 in place of r into $V = (\frac{4}{3})\pi r^3$.

4,187 cubic centimeters

Found $r = 5$ correctly but then doubled it to a diameter of 10 before plugging into the volume formula, confusing radius with diameter.

1,570 cubic centimeters

Found $r = 5$ correctly but used $V = 4\pi r^3$ (the surface-area coefficient of 4) instead of $V = (\frac{4}{3})\pi r^3$, dropping the $\frac{1}{3}$ factor.

Shortcut: Once r is known, note that $V = (r/3) \times SA$, since $(\frac{4}{3})\pi r^3 = (r/3)(4\pi r^2)$; plugging in gives $V = (\frac{5}{3})(314) \approx 523.3$ without recomputing r from scratch.

Desmos Tip: Use the calculator's solver to find r from $4(3.14)r^2 = 314$, then compute $(\frac{4}{3})(3.14)r^3$ with that r in a second line.